

August 28, 2026

1 Commentary: latent fibers and adaptive polyhedral closure

The homogeneous predecessor studied above is only one prefix-compatible causal mechanism. In particular, even a proof that

$$m_n \asymp \sqrt{n}$$

would not contradict the possibility that the full causal zero-repair problem satisfies

$$\mathcal{B}_n = O(\log n).$$

Rather, it would show that a genuinely non-homogeneous, multiscale mechanism is required.

This section records one possible structural interpretation of such a mechanism. The point is not to assert a theorem about the asymptotic growth of $\mathcal{B}_n$, but to identify a finite-dimensional polyhedral object that may encode the missing freedom.

1.1 Continuation viability as a zero set

Fix a target law $B_\star$ and a finite support cutoff

$$\Delta_M = \left\{ a = (a_0, \dots, a_M) : a_j \geq 0, \sum_{j=0}^M a_j = 1 \right\}.$$

Let

$$\mathcal{C}_r(B_\star)$$

denote the laws in Δ_M admitting an exact depth- r zero-repair continuation all of whose terminal laws stochastically dominate $B_\star$.

For the concrete three-atom target

$$B_\star = c_0 \delta_0 + (c_1 - c_0) \delta_1 + (1 - c_1) \delta_2,$$

the terminal domination condition is exactly

$$b_0 \leq c_0, \quad b_0 + b_1 \leq c_1.$$

Define the nonnegative terminal defect

$$d_0(b) = \max \{0, b_0 - c_0, b_0 + b_1 - c_1\}.$$

For $r \geq 0$, define recursively

$$d_{r+1}(a) = \min_{b^{(0)}, \dots, b^{(4)} \in \Delta_M} \left[\sum_{x=0}^4 P(x) d_r(b^{(x)}) + \lambda \left\| Q * a - \sum_{x=0}^4 P(x) \delta_x * b^{(x)} \right\|_1 \right], \quad (33)$$

where

$$\lambda > 0.$$

Every term in the objective is nonnegative. Consequently,

$$\boxed{d_r(a) = 0 \iff a \in \mathcal{C}_r(B_\star).} \quad (34)$$

Indeed, zero defect forces the zero-repair residual to vanish exactly and forces every child defect to vanish. The converse follows by choosing an exact feasible continuation.

Thus the complete depth- r causal viability condition is represented by a single scalar zero set.

For fixed finite M , each d_r is convex and polyhedral, hence admits a representation

$$d_r(a) = \max_{\alpha \in \mathcal{A}_r} \left\{ g_{r,\alpha}^\top a - h_{r,\alpha} \right\}. \quad (35)$$

The exponentially large history tree has therefore been projected onto a finite-dimensional polyhedral value function.

1.2 Adaptive separation is exact

Suppose only a subset

$$\mathcal{H}_r \subseteq \mathcal{A}_r$$

of the affine pieces is inserted explicitly into the current linear program. Call this subset the *head*; the remaining valid cuts form the *tail*.

For a candidate law a , define

$$c_\alpha(a) = g_{r,\alpha}^\top a - h_{r,\alpha}.$$

After solving the reduced problem using only $\mathcal{H}_r$, scan the complete tail and compute

$$\max_{\alpha \in \mathcal{A}_r \setminus \mathcal{H}_r} c_\alpha(a).$$

Any violated tail cuts are promoted to the head and the reduced problem is solved again.

Proposition 1 (Exact adaptive closure). *Assume that every affine inequality in $\mathcal{A}_r$ is valid and that the tail separation step checks every inactive cut exactly. If the adaptive procedure terminates at a point $a_\star$ for which no inactive cut is violated, then $a_\star$ is feasible for the full polyhedral problem.*

Proof. At termination, all head inequalities hold because $a_\star$ solves the reduced problem. Every tail inequality also holds by the exhaustive separation test. Hence every inequality in $\mathcal{A}_r$ holds. $\square$

Thus a head–tail implementation need not discard any information. It changes only which facets are present explicitly in the LP at a given moment.

Schematically,

$$\boxed{\text{small active LP} \longrightarrow \text{full separation} \longrightarrow \text{promote violations} \longrightarrow \text{repeat}.} \quad (36)$$

1.3 Effective active-face complexity

Problem 3 asks whether the number and total cost of active-set transitions can be controlled.

The total number of stored facets may, however, substantially overstate the local geometric complexity. At a point a , define

$$\widehat{d}_r(a) = \max_{\alpha \in \mathcal{A}_r} c_\alpha(a)$$

and the facet gaps

$$\Delta_\alpha(a) = \hat{d}_r(a) - c_\alpha(a) \geq 0.$$

For a scale parameter $\tau > 0$, introduce

$$w_\alpha(a) = \exp\left(-\frac{\Delta_\alpha(a)}{\tau}\right), \quad p_\alpha(a) = \frac{w_\alpha(a)}{\sum_\beta w_\beta(a)}.$$

Define the effective active-face count

$$N_{\text{eff}}(a) = \frac{1}{\sum_\alpha p_\alpha(a)^2}. \quad (37)$$

This quantity satisfies

$$1 \leq N_{\text{eff}}(a) \leq |\mathcal{A}_r|.$$

If

$$N_{\text{eff}}(a) \ll |\mathcal{A}_r|,$$

then the local geometry is head-dominated: only a small number of facets lie near the active envelope.

If instead

$$N_{\text{eff}}(a) \asymp |\mathcal{A}_r|,$$

the geometry is diffuse and a small active-set representation may offer little compression.

Equation (37) is only a diagnostic; it does not alter the feasible set. A geometrically adaptive implementation may therefore choose between a full-pool LP and a head-tail LP without affecting correctness, provided exhaustive separation is retained.

1.4 Bellman fibers

There is a second, more structural source of compression.

For a parent law a , define its exact one-step Bellman fiber

$$\mathfrak{F}(a) = \left\{ (b^{(0)}, \dots, b^{(4)}) \in (\Delta_M)^5 : Q * a = \sum_{x=0}^4 P(x) \delta_x * b^{(x)} \right\}. \quad (38)$$

The parent law a is therefore only the boundary coordinate of a much larger latent set of child decompositions.

Different points of $\mathfrak{F}(a)$ may induce the same parent law and the same continuation value. The projection

$$(b^{(0)}, \dots, b^{(4)}) \mapsto a$$

can therefore hide substantial latent geometric freedom.

This suggests a continuous analogue of a fiber multiplicity: the computational complexity of the explicit history tree may be much larger than the complexity of its projected root viability polytope.

Unlike the finite combinatorial setting, the fibers in (38) are continuous polytopes, so there is no canonical counting entropy. Natural candidates for a geometric “reserve” include

$$\dim \mathfrak{F}(a),$$

relative polytope volume, normal-cone complexity, or the number of distinct child decompositions exposed by relevant Bellman dual directions.

At present this is a heuristic structural analogy, not an entropy theorem.

1.5 A possible multiscale interpretation

The homogeneous predecessor selects one law

$$B_{n+1} = D(B_n)$$

at every step.

The full causal problem has substantially more freedom: at each node it may choose a different point in the corresponding Bellman fiber, depending on the complete history.

Thus the distinction may be written schematically as

$$\boxed{\text{homogeneous predecessor} = \text{one selected trajectory through law space,}}$$

whereas

$$\boxed{\text{general causal continuation} = \text{history-dependent navigation through Bellman fibers.}}$$

This offers a possible explanation for the coexistence of

$$m_n \gg \log n$$

along the homogeneous predecessor orbit and

$$\mathcal{B}_n = O(\log n)$$

for the unrestricted causal problem.

The missing logarithmic mechanism, if it exists, may therefore reside not in a better approximation to the homogeneous predecessor trajectory, but in repeated exploitation of latent continuation fibers across multiple scales.

1.6 Refined complexity questions

The preceding viewpoint suggests the following additions to Problems 3–5.

Problem 6 (Effective facet complexity). Can one bound, uniformly over the relevant Bellman trajectory, the effective number of near-active facets

$$N_{\text{eff}}(a)$$

or an equivalent normal-cone complexity?

In particular, can one prove a scale-wise bound that is independent of the total number of stored cuts?

Problem 7 (Fiber complexity). For viable parents a , can the geometry of

$$\mathfrak{F}(a)$$

be quantified in a manner that controls the cost of extending the zero-repair tree?

Is there a monotone or telescoping notion of latent continuation reserve?

Problem 8 (Bounded doubling premium). Let

$$D_r(B)$$

denote the minimum mean of a root admitting a depth- r exact zero-repair tree whose leaves all stochastically dominate B .

Can one find a prefix-compatible sequence B_{2^k} and a universal constant $C < \infty$ such that

$$D_{2^k}(B_{2^k}) - \mathbb{E}B_{2^k} \leq C$$

at every dyadic scale?

Such a bound would produce an $O(1)$ surcharge per dyadic scale and is therefore structurally compatible with

$$\mathcal{B}_n = O(\log n).$$

1.7 Interpretation

The obstacle formulation of Sections 4–7 identifies the cost generated when a quartic forcing is repeatedly projected back into the monotone probability cone.

The Bellman-fiber formulation emphasizes a different source of structure: the full causal problem does not follow a single projected trajectory. It retains a latent family of continuation choices at every node.

Accordingly, two distinct questions should be separated:

How costly is the homogeneous reflected trajectory?
How much can history-dependent fiber freedom reduce that cost?

The first is the quartic predecessor problem stated above.

The second may be the relevant question for the matching upper bound

$$\mathcal{B}_n = O(\log n).$$

The two problems are related, but they need not have the same scaling law.